

Chapter 138 — Urothelial Tumours of the Upper Urinary Tract and Ureter — Chapter Notes

INCIDENCE

- Urothelial carcinomas are the **sixth most common tumours** in developed countries; **upper tract urothelial carcinoma (UTUC)** accounts for only **5–10% of urothelial tumours**.
- **Annual incidence in Western countries** $\approx$ **2 per 100,000**; the Surveillance, Epidemiology and End Results (SEER) database gives an annual incidence of roughly 1.9–2.1 per 100,000 person-years.
- **Overall incidence is rising**, driven by **ureteric tumours** (rising) while **pelvicalyceal tumours are static or marginally falling**.
- Reasons for the rise: **better diagnostic methods** and **longer survival after bladder cancer surgery** (more patients live long enough to develop an upper tract tumour).
- Stage and grade migration: **proportion of early-stage tumours has risen markedly** over three decades, and the **proportion of high-grade UTUC has also risen**.
- **Peak incidence: age 70–90, males, and the Balkan countries, where UTUC is 40% of all renal neoplasms**.

Distribution and multiplicity

- **Usually a single renal unit; bilateral in up to 5%.**
- **Pelvicalyceal tumours about twice as common as ureteric tumours** (elsewhere in the chapter stated as only slightly more common).
- **Multifocal in 10–20%; concomitant carcinoma in situ (CIS) in 11–36%; concurrent bladder cancer in 17%.**
- **Prior bladder cancer is far commoner in American than in Chinese males with UTUC; Asian patients present with higher-grade disease.**
- **At diagnosis, $\approx$ 60% are invasive and $\approx$ 7–9% are metastatic.**
- **The absence of a thick muscularis propria** in the upper tract explains **earlier upstaging** and the high proportion of invasive tumours at presentation.

Recurrence after treatment, and UTUC after bladder cancer

- After UTUC treatment, recurrence involves the **bladder in 22–47%** and the **contralateral upper tract in 2–6%**.
- **UTUC incidence is increased after treated bladder cancer: $\approx$ 7.5% after bladder-sparing surgery plus intravesical bacillus Calmette–Guérin (BCG) versus 3–5% after radical cystectomy for muscle-invasive disease.**

SEX, RACE, AGE VARIATIONS, AND MORTALITY RATE

- **Male : female $\approx$ 2 : 1; Black individuals affected about twice as often as White individuals.**
- **UTUC patients are older than bladder cancer patients and have a worse prognosis.**
- **Disease mortality rises with age, male sex, Black non-Hispanic race and advanced stage;** higher grade (III–IV vs I–II) and stage also raise mortality in recent SEER data.
- **Five-year disease-specific survival falls steeply with stage:** in situ $\approx$ 95%, localised $\approx$ 89%, regional $\approx$ 63%, distant $\approx$ 17% (overall $\approx$ 75%).

SEX AND RACE — THE CONFLICTING EVIDENCE

Question	Population-based registries (SEER)	Multicentre series, adjusted for clinicopathological features
Sex	Higher mortality in women; women more often present with pT3 disease	Sex not an independent predictor of pathology, survival or disease-specific mortality after radical nephroureterectomy
Race	Worse outcome in Black patients and in non-White races	No outcome difference by race (including Japanese vs European)

- **Reason for the discordance:** registry data are unadjusted and reflect presentation stage; once stage and grade are controlled, the demographic effect largely disappears.

SOMATIC AND GERMLINE ALTERATIONS: GENOMIC CLASSIFICATION AND FAMILIAL PREDISPOSITION

- **UTUC and bladder cancer are genetically distinct;** when one patient has both, the tumours are usually clonally related.
- **Five molecular subtypes:** hypermutated, TP53/MDM2-mutated, RAS-mutated, FGFR3-mutated and triple-negative (no TP53/MDM2, RAS or FGFR3 mutation). The **KRAS/RAS-mutated group has the worst prognosis.**
- **Higher-stage disease: fewer RTK/RAS alterations, more TP53/MDM2 alterations.** Molecular classification may in future direct targeted therapy.
- **Germline analysis** serves two purposes: estimating the risk of subsequent bladder cancer and identifying Lynch syndrome (LS).
- The **Amsterdam criteria** identify families likely to carry LS. Immunohistochemistry of LS-related UTUC shows **loss of the protein of the mutated mismatch repair (MMR) gene** in the vast majority; **MSH2 carriers have the greatest tumour tendency.**
- A **germline MMR mutation is found in about 5–9%** of screened UTUC patients.
- **Screen for LS:** all UTUC patients under 60 and anyone with a family history of UTUC; the most recent recommendation extends **molecular tumour testing to patients under 65** or with a personal or family history of LS-related cancer, followed by germline testing.

FIGURE 138.1 SELECTION OF PATIENTS WITH UTUC FOR HEREDITARY SCREENING DURING THE FIRST MEDICAL INTERVIEW

Adapted from the European Association of Urology (EAU) guidelines on upper urinary tract urothelial carcinoma, 2020 update.

- **Systematic screening at the first interview** → suspicion of hereditary UTUC if any of: **age < 60; personal history of an LS-spectrum cancer; first-degree relative < 50 years with an LS-spectrum cancer; two first-degree relatives with an LS-spectrum cancer.**
- Suspected hereditary → **germline DNA sequencing** (mutation found in 5–9%) → **evaluate other LS cancers** (colorectal, gastrointestinal, endometrial, ovarian, skin) → **close monitoring** → **familial genetic counselling.**
- A **sporadic-appearing UTUC with abnormal MMR immunohistochemistry** (MLH1, MSH2, MSH6, PMS2) should also proceed to germline sequencing.

KEY POINTS: EPIDEMIOLOGY

- UTUC makes up 5–10% of urothelial cancers.
- Highest incidence at age 70–90, and in the Balkan countries, where UTUC is 40% of all renal neoplasms.

- Multifocal UTUC in 10–20%; concurrent bladder cancer in 17%.
- UTUC is twice as frequent in males as in females.
- All patients under 60 with UTUC, and anyone with a family history of UTUC, should be screened for LS.

Concept — *UTUC is rare, rising and presents late: the thin upper tract wall lets tumours invade early, so the epidemiology is dominated by invasive disease, by the bladder as the commonest site of recurrence, and by a small hereditary (Lynch) group that must be actively sought in the young.*

RISK FACTORS

Genetic Predisposition

- **Familial/hereditary UTUC = hereditary nonpolyposis colorectal carcinoma (HNPCC), i.e. Lynch syndrome**, caused by mutations in the **MMR genes MLH1, MSH2, MSH6 and PMS2**.
- **LS cancers: colon, urothelial, gastric, pancreatic, uterine, sebaceous, ovarian**. The urothelial cancers are **mainly upper tract**; bladder involvement is uncertain and a bladder tumour in LS may simply be **seeding from the upper tract**.
- **LS confers ≈ 14-fold UTUC risk** over the general population, rising to **≈ 75-fold in MSH2 carriers**.
- **LS-associated UTUC: younger (< 60; mean ≈ 55) and with relatively more women** than sporadic UTUC.
- Suspect genetic predisposition in **young patients, a personal history of LS cancer, or two first-degree relatives with an LS cancer** (especially colon or endometrium) → short structured interview (Figure 138.1) → **DNA sequencing and family counselling** in high-risk cases.

Environmental Factors

Tobacco Exposure

- **Smoking is the most important modifiable risk factor**; it raises UTUC risk **2.5- to 7-fold**.
- Mechanism: **aromatic amines** (arylamines, benzopyrene, dimethylbenzanthracene) are activated to **mutagenic N-hydroxylamines**; detoxification by **cytochrome P450 (CYP1A1), glutathione S-transferases and N-acetyltransferases** is modulated by genetic polymorphisms.
- **Dose-response** with cigarettes per day and years smoked — risk roughly triples from ≤ 20 to ≥ 60 pack-years.
- **Cessation for > 10 years reduces risk by 60–70%**; former smokers carry about half the risk of current smokers.
- **The ureter is at greater smoking-related risk than the pelvicalyceal system**.

Coffee, Tea, and Yerba Mate

- **Coffee and tea: no significant association** with UTUC; any apparent coffee effect is **confounded by smoking**. Water, coffee, tea and dairy intake were unrelated to urothelial cancer in a large prospective European cohort.
- **Yerba mate** (South America) supplies **polycyclic aromatic hydrocarbons** and roughly doubles urothelial cancer risk, but has **never been studied specifically for UTUC**.

Alcohol Consumption

- **Ever-drinkers have a modestly higher UTUC risk** than never-drinkers, with a **dose-response**; the risk threshold is **≥ 15 g of alcohol daily**.

Occupation

- **"Amino tumours"**: UTUC from occupational **aromatic amines** — dyes, textiles, rubber, chemicals, petrochemicals, coal. **Benzidine and β -naphthylamine** are the implicated amines, **banned since the 1960s** in most industrialised countries.
- **Average exposure needed $\approx$ 7 years; latency up to 20 years** after exposure ends.
- Raised risk also with the **chemical, petroleum and plastics industries, coal or coke**, and highest with **asphalt or tar; polycyclic aromatic hydrocarbons and nitrosamines** (dose effect); **chlorinated solvents** (trichloroethylene, tetrachloroethylene) in metallurgy and printing.

Aristolochic Acid Nephropathy

- **Balkan endemic nephropathy (BEN)**: UTUC incidence **60–100 times higher** in some rural Balkan areas; the excess fell sharply (from $\approx$ 57-fold to $\approx$ 11-fold) between 1988 and 1998.
- BEN is a **proximal tubular disorder with low-molecular-weight proteinuria and dense interstitial fibrosis**.
- **Ochratoxin A** was investigated for decades; it is carcinogenic in animals but its lesions **do not match BEN tumours** histologically.
- **Chinese herb nephropathy**: Belgium, 1992–1993 — end-stage renal failure after slimming herbal products in which *Stephania tetrandra* was replaced in error by *Aristolochia fangchi*; **nearly half later developed UTUC**, histologically and genetically identical to BEN tumours.
- **Aristolochic acid** (from *A. fangchi* and *A. clematitis*, a weed of wheat fields) is the common cause. BEN is **familial but not inherited** — a shared **dietary exposure**; **bladder cancer is not increased** in affected families.
- Mechanism: the metabolite **aristolactam forms aristolactam–deoxyadenosine DNA adducts** (lasting exposure biomarkers) $\rightarrow$ **A:T $\rightarrow$ T:A transversions**, mainly on the non-transcribed strand, including **TP53 codon 139**. This signature mutation predominates in BEN and Chinese herb nephropathy.
- **Fewer than 10% of exposed people develop UTUC**.
- High UTUC prevalence in **Taiwan and China** is linked to aristolochia-containing herbal remedies.
- **Worse prognosis in aristolochic acid UTUC**: female sex, **tumour > 3 cm, stage T3–T4**.

Analgesics

- **Phenacetin** first implicated in 1961; carcinogenesis mainly **indirect, through nephrotoxicity and papillary necrosis** (the confirmed mechanism) rather than direct mutagenesis. **Withdrawn about 30 years ago and replaced by paracetamol**; phenacetin-related UTUC is declining. **Mean latency $\approx$ 22 years**.
- **Analgesic abuse is an established risk factor**: history of abuse in **22% of pelvic and 11% of ureteric tumours**. Most abusers take **combinations** (caffeine, codeine, paracetamol, salicylates).
- **Pathognomonic histology**: **basement membrane thickening (capillarosclerosis) and papillary scarring** — present in up to 15% of UTUC; its presence should raise suspicion of analgesic abuse and of **contralateral disease**.
- **Tumour grade correlates with the degree of papillary scarring**; scarring is not linked to squamous metaplasia.

Arsenic

- **Inorganic arsenic in artesian well water** (south-west coast of Taiwan, Chile) is associated with UTUC; up to a quarter of Taiwan's UTUC arises in this region.
- Chronic exposure causes **blackfoot disease** — a peripheral vascular disease with **dry gangrene of the extremities**.

- **Features of UTUC in blackfoot-endemic areas: women affected twice as often** (arsenic fumes during cooking — inhalation plus ingestion), **younger onset (55–60)**, **ureter involved twice as often as the pelvis**.
- The causal link is **not established for UTUC** (it is stronger for bladder cancer): many patients never have blackfoot disease, and north-east Taiwan has a high UTUC rate without it.

Chronic Inflammation and Infection

- **Chronic urinary infection** weakens the urothelium; stasis and urinary irritants promote carcinogenesis; risk raised $\approx$ 1.5–2-fold.
- **Urolithiasis history in 5–8% of UTUC**, but tumour incidence in stone formers remains low; risk higher in women, with renal pelvic calcification, and after stone-related admission.
- **Chronic infection + stones + obstruction → squamous cell carcinoma** (less often adenocarcinoma).

Iatrogenic Factors

- **Alkylating chemotherapy (cyclophosphamide) → acrolein → UTUC**, with epidermoid pathology on chronic exposure.
- **Anthranoid and chemical laxatives used > 1 year**: strong association; mechanism unknown.
- **Pelvic radiotherapy for gynaecological cancer**: raised risk, but **lower than after chemotherapy**.

ENDEMIC AND EXPOSURE-RELATED UTUC COMPARED

Feature	Aristolochic acid (BEN, Chinese herb nephropathy)	Arsenic (blackfoot disease)	Analgesic abuse (phenacetin)	Lynch syndrome
Geography / setting	Rural Balkans; Belgium 1992–93; Taiwan, China	South-west Taiwan, Chile	Chronic analgesic users	Families with MMR mutation
Mechanism	Aristolactam adducts → A:T → T:A transversion (TP53)	Well-water arsenic; causal link unproven for UTUC	Papillary necrosis ; capillarosclerosis	MMR deficiency (MSH2 highest risk)
Sex	No sex predominance	Female 2 : 1	—	Relatively more women
Age	$\approx$ 10 years earlier than sporadic	55–60	Long latency ($\approx$ 22 years for phenacetin)	< 60
Site	Bilateral more often	Ureter 2 : 1 over pelvis	Pelvis > ureter; think contralateral	Upper tract > bladder
Inherited?	Familial, not inherited	No	No	Inherited (autosomal dominant)

- All four share a **greater tendency to multiple and bilateral recurrence** than sporadic UTUC. **Trap**: "familial" in BEN reflects shared diet, not a germline mutation.

KEY POINTS: RISK FACTORS

- Familial/hereditary UTUC is linked to HNPCC (Lynch syndrome).
- Tobacco raises the relative risk of UTUC from 2.5 to 7.
- UTUC "amino tumours" follow occupational exposure to carcinogenic aromatic amines.
- Balkan nephropathy is familial, not inherited, and follows dietary exposure to aristolochic acid.
- A history of analgesic abuse is present in 22% of pelvic and 11% of ureteric tumours.

- Excess inorganic arsenic in artesian well water is a significant risk factor for UTUC.

Concept — *Almost every UTUC risk factor is a urine-borne carcinogen concentrated in the upper tract (tobacco amines, aristolochic acid, arsenic, phenacetin metabolites, acrolein); the exposure-driven forms behave as a field disease — earlier, bilateral and multiply recurrent — which is why they bear directly on treatment choice.*

HISTOPATHOLOGY

- **> 90% of upper tract tumours are urothelial**; pure non-urothelial cancers (squamous cell carcinoma, adenocarcinoma) are rare.
- **Histological variants in ≈ 25%**; they correspond to **high-grade tumours with a worse prognosis**.

Normal Upper Tract Urothelium

	Renal pelvis and ureter	Bladder
Embryological origin	Mesoderm	Endoderm
Urothelium	Similar; relatively abundant in pelvis and calyces; continuous from calyces to distal ureter, possibly into collecting ducts	Similar
Muscle layer	Much thinner	Thick muscularis propria

- Possible **extension of urothelium into the collecting ducts** could justify urothelial-cancer regimens for collecting duct carcinoma.

Renal Pelvis and Calyces

- **Transitional epithelium over two muscle layers plus fibrous tissue**; the muscle is **spiral/helical**, originating in the minor calyces.

Ureter

- **Two continuous thin spiral layers**: outer layer tightly spiralled, inner layer looser; a **third outer longitudinal layer in the lower third**.
- These **merge with the three bladder layers** (inner longitudinal, middle circular, outer longitudinal). The **serosa (adventitia)** of loose connective tissue carries vessels and lymphatics.

Urothelial Neoplasms

Benign Lesions: Papillomas and von Brunn Nests

- **Papillomas, inverted papillomas and von Brunn nests are benign**, but synchronous or metachronous UTUC cannot be excluded → **close surveillance**.
- **Malignancy found in ≈ 18% of ureteric inverted papillomas**. **Type 1 benign**; **type 2 potentially malignant** — but the two **cannot be told apart**, so **all inverted papillomas are followed for at least 2 years**.

Metaplastic and Dysplastic Lesions

- UTUC often develops through **hyperplasia → dysplasia → CIS**.
- **CIS spreads in patches, up to the collecting ducts**. **Severe dysplasia predicts a worse prognosis** through higher recurrence in the distal ureter and bladder.

UROTHELIAL MALIGNANT LESIONS

Urothelial Carcinoma

- **> 90% of upper tract tumours.** Morphology as in the bladder: **flat (CIS), papillary or sessile; unifocal or multifocal.**
- Histologically like bladder cancer, but the **thin muscle allows earlier muscle invasion**, and invasion of **renal parenchyma or periureteric adventitia** is more likely than in the bladder.
- **CIS is hard to diagnose:** it may look like a **whitish plaque, epithelial hyperplasia or a smooth red patch** (increased submucosal vascularity).
- **Collecting duct carcinoma** shares features through common embryology but is **classed as kidney cancer, not UTUC.**

FIGURES 138.2 AND 138.3 HISTOLOGY

- **Figure 138.2A:** high-grade urothelial carcinoma, showing the **urothelial layer, lamina propria and muscle**, with a focus of **lamina propria invasion**. **Figure 138.2B:** CIS — carcinoma confined to the urothelium, **no underlying infiltration**.
- **Figure 138.3A:** low-grade carcinoma — **cell orientation preserved**. **Figure 138.3B:** CIS — **loss of orientation**, many mitoses, apoptosis.

Micropapillary Variant of Urothelial Carcinoma

- **Variants (up to 25%):** squamous, glandular, sarcomatoid, micropapillary, neuroendocrine, lymphoepithelial — all linked to aggressive behaviour, although **not independently prognostic once other clinicopathological features are adjusted for.**
- **Micropapillary UTUC** is rare and usually **advanced at presentation** (most pT3, frequent CIS and nodal involvement); it is an **independent predictor of worse progression-free and cancer-specific survival.**

Nonurothelial Malignant Lesions

- Rare, ranging from benign to highly malignant; **squamous cell carcinoma and adenocarcinoma are the commonest.**

Tumour	Frequency	Associations	Behaviour / treatment
Squamous cell carcinoma (SCC)	0.7–7% of upper tract cancers; 6× commoner in pelvis than ureter	Analgesic abuse, chronic inflammation; squamous carcinoma in calcified papillae after analgesic abuse	Moderately-poorly differentiated; invasive at presentation
Adenocarcinoma	< 1% of renal pelvic tumours	Calculi, long-term obstruction, inflammation	Advanced at diagnosis, poor prognosis
Fibroepithelial polyp, neurofibroma	Rare	—	Benign; simple excision
Neuroendocrine, haematopoietic, sarcoma	Very rare	—	Excision plus adjuvant therapy, modelled on the same histology elsewhere

KEY POINTS: HISTOPATHOLOGY

- Urothelial carcinomas are > 90% of upper tract tumours; pure non-urothelial cancers are rare.
- Variants occur in ≈ 25% of UTUC.
- Papillomas, inverted papillomas and von Brunn nests are usually benign.
- UTUC often progresses hyperplasia → dysplasia → CIS.

- CIS is difficult to diagnose because of wide morphological variation.
- Invasion into muscle, renal parenchyma or adventitia is more likely in the upper tract.

Concept — *Upper tract and bladder urothelium look the same but sit on different walls: the thin mesodermal muscle is the single fact that explains earlier invasion, the dominance of invasive disease and the limited value of superficial biopsy in staging.*

DIAGNOSIS

Symptoms and signs

- **Haematuria is the commonest sign (56–98%)**, gross or microscopic. Yield of UTUC is low: $\approx 0.75\%$ of gross and $\approx 0.17\%$ of microscopic haematuria.
- **Flank pain is second (20–32%)**: usually **dull** from gradual obstruction and hydronephrosis; **colic when clots pass**. Flank pain is **commoner than with bladder tumours** (10–40%).
- **Palpable lumbar mass $\approx 10\%$; asymptomatic, incidental $\approx 15\%$.**
- **Systemic symptoms** (anorexia, weight loss, fatigue, fever, bone pain, night sweats, abdominal or flank mass) → **immediate metastatic work-up**; they carry a worse prognosis.
- UTUC is a **rare autopsy finding** — most are diagnosed in life.

Gross features

- **Ureteric tumours are low grade and low stage in 55–75%.**
- **85% papillary, 15% sessile. Invasion (T1–T2) in 50% of papillary vs $> 80\%$ of sessile tumours.**
- **Invasive tumours are commoner in the ureter than in the bladder; renal pelvic tumours invade lamina propria or muscle in 50–60%.**

Endoscopy of the Lower Tract

Cystoscopy

- **Cystoscopy always** — UTUC frequently coexists with bladder cancer.
- **Positive cytology + negative cystoscopy (including urethra) → evaluate the upper tracts.**

Endoscopy of the Upper Tract

Ureterorenoscopic Evaluation

- **Flexible ureteroscopy (URS) is the key tool** for visualising ureter, pelvis and collecting system, and for biopsy.
- **Adding URS raises diagnostic accuracy from 75–85% (excretory or retrograde urography alone) to $\approx 90\%$.**
- **Pyelovenous/pyelolymphatic dissemination** from URS was reported historically but **not confirmed with modern technique** → URS whenever needed.
- **Biopsy (forceps or brush) determines grade in $\approx 90\%$, with a low false-negative rate regardless of sample size; biopsy–specimen correlation 78–92%. Visual grading alone is right in $\approx 70\%$ → biopsy needed.**
- **Undergrading and understaging are common**: specimens are small and rarely include deep layers; loop biopsies contain lamina propria far more often than cup biopsies; a Ta biopsy is frequently upstaged to T3 at resection. A grade 3 biopsy predicts stage accurately in $> 90\%$.
- Hence, EAU: **stage cannot be assessed from URS biopsy**; stratify by grade + visual appearance + imaging (hydronephrosis) + cytology, and follow kidney-sparing patients intensively.

- URS allows **selective in-situ cytology** sampling.
- **Indications for diagnostic URS:** diagnostic uncertainty; kidney-sparing treatment being considered; solitary kidney.
- **URS before radical nephroureterectomy (RNU) increases intravesical recurrence**, particularly with renal pelvic tumours and with biopsy, without affecting long-term survival. Whether an **immediate intravesical instillation after URS with biopsy or laser treatment** prevents this is under study.
- **Enhanced imaging:** narrow-band imaging is promising but preliminary; **optical coherence tomography and confocal endomicroscopy (Cellvizio)** correlate well with final histology for grade and stage.

FIGURE 138.4 URETEROSCOPIC APPEARANCES

- **A:** multiple carcinomas in the upper calyceal group — **scrutinise all urothelium** to avoid missed sites. **B:** same lesions with **endoscopic filters**, which help reveal small tumours. **C:** sessile tumour with a papillary surface. **D:** the same under filters.

Antegrade Endoscopic Evaluation

- **Percutaneous nephroscopy, antegrade URS and urography** allow visualisation, biopsy and **resection/debulking of pelvic tumours with larger scopes**.
- **EAU:** consider for **low-risk renal pelvic UTUC inaccessible or difficult with flexible URS**. Largely **abandoned** with better flexible scopes.
- **Risk:** **tumour seeding of the retroperitoneum and nephrostomy tract**.

Radiologic Imaging

- **Computed tomography urography (CTU) has the highest diagnostic accuracy** (pooled sensitivity $\approx 92\%$, specificity $\approx 95\%$). Thin isotropic sections allow multiplanar reconstruction; accuracy equals intravenous pyelography with less labour; drawback is **high radiation dose**.
- **Suggestive signs:** **radiolucent filling defect** (seen in about 50–75%), **obstruction or incomplete filling, non-visualisation of the collecting system**.
- **Differential of a filling defect:** stone, blood clot, bowel gas, extrinsic compression, sloughed papilla, fungus ball. Calcification on ultrasound or unenhanced CT identifies stones.

Density on CT	Mean	Range
Urothelial carcinoma	46 HU	10–70 HU
Radiolucent uric acid stone	100 HU	80–250 HU

- **Hounsfield unit (HU) measurement distinguishes the two commonest radiolucent filling defects**.
- **Hydronephrosis**, with or without a filling defect, indicates **invasive ureteric tumour in up to 80%** and is linked to **advanced disease and poor outcome**; it is an **independent predictor**.
- **Non-visualisation is not clearly linked to stage:** seen in only $\approx 20\%$ of pelvic tumours and $\approx 37\text{--}45\%$ of ureteric tumours.
- **Flat lesions without mass effect are generally invisible on CT**. CT shows extraluminal masses, nodes and metastases; **pathological nodes predict non-organ-confined disease**. CT has **poor sensitivity for invasion of small lesions** and predicts TNM stage correctly in only a majority (both under- and overstaging). It also identifies poorly functioning, obstructed segments.

IMAGING MODALITIES COMPARED

Modality	Role	Key limitation or point
CT urography	First-line for diagnosis and staging	Radiation; misses flat lesions; weak for invasion of small tumours

Modality	Role	Key limitation or point
Magnetic resonance (MR) urography	When CT radiation or iodinated contrast is contraindicated	Sensitivity $\approx$ 75% for tumours < 2 cm; limit gadolinium if creatinine clearance < 30 mL/min (nephrogenic systemic fibrosis)
Retrograde urography	Radiolucent non-calcified lesions; poorly filled or obstructed calyces	Accuracy $\approx$ 75%; often combined with URS
Ultrasonography	Painless gross haematuria	Some report sensitivity equal to urography
F-fluorodeoxyglucose positron emission tomography/CT (FDG-PET/CT)	Nodal/metastatic assessment when iodinated contrast is contraindicated	Suspicious nodes predict worse recurrence-free survival
Split-function renal scan	Contribution of the diseased versus contralateral kidney	Contralateral assessment is mandatory (bilateral disease, function)

- EAU: CT urography is preferred to MR urography for diagnosis and staging.

Cytology and Tumor Markers

- Overall urine cytology sensitivity $\approx$ 65%, strongly grade-dependent: $\approx$ 20% grade 1, $\approx$ 45% grade 2, $\approx$ 75% grade 3.
- Voided cytology gives no site of origin; it is less sensitive for UTUC than for bladder cancer. Selective sampling or ureteric washing by catheter (in situ in the collecting system) is more accurate; saline washings raise the cell yield. False results remain common.
- Brush biopsy + cytology raises sensitivity to $\approx$ 90% (high for low grade, near-complete for high grade), at the cost of haemorrhage and perforation.
- Take cytology before any contrast — ionic high-osmolar agents degrade the specimen.
- Fluorescence in situ hybridisation (FISH) is as sensitive as in bladder cancer; best suited to low-grade disease under surveillance after kidney-sparing therapy, but not routinely used.

KEY POINTS: DIAGNOSIS

- Localised disease: haematuria, dysuria and flank pain.
- Advanced disease: flank or abdominal mass, weight loss, anorexia and bone pain.
- Cystoscopy should always be performed.
- Flexible URS with biopsy is the key diagnostic approach.
- CT urography has the highest diagnostic accuracy.
- Urine collection or ureteric washing via a ureteric catheter gives the most accurate cytology.

Concept — Every UTUC test is good at grade and poor at stage: biopsy is shallow, CT misses flat and small invasive lesions, and cytology tracks grade — so stage is inferred from a composite (grade, appearance, hydronephrosis, cytology), and that composite drives the choice between kidney-sparing treatment and RNU.

STAGING AND CLASSIFICATION

Natural History

Recurrence

- Older series: **metastases at diagnosis in up to 19%**. Contemporary matched cohorts: **stage-for-stage cancer-specific outcomes are similar** for upper and lower tract cancers despite UTUC's more invasive, less differentiated character.

Stage after radical surgery	Which does worse	Why
Non-muscle-invasive	Bladder cancer	Bladder cancers reaching cystectomy are self-selected aggressive (lymphovascular invasion, persistence after transurethral resection, BCG failure); UTUC goes to RNU early because it cannot be staged
pT2 and pT3	No difference	—
pT4	UTUC	Bladder pT4a includes prostatic stromal invasion, still curable by cystoprostatectomy

Involvement of the Ureter or Renal Collecting System

- **Ureteric tumours: 70% distal, 25% mid, 5% proximal** — attributed to **downstream implantation**.
- **Synchronous or metachronous bilateral disease: 1.6–6% of sporadic UTUC.**
- **Stage-adjusted, ureteric and/or multifocal tumours do worse than renal pelvic tumours.**

Synchronous and Asynchronous Localizations

Upper tract recurrence after bladder cancer.

- **Incidence 0.75–6%, late** (median > 3 years); traditional series **2–4% over 17–170 months**, some up to 25% (selection, grade, stage and dysplasia explain the spread). **After cystectomy ≈ 0.75–6.4%** in the largest meta-analysis.
- **Risk factors:** tumour at the **distal ureter/ureteric orifice**, **multifocal or recurrent** bladder cancer, **non-muscle-invasive bladder cancer**, **CIS**; risk rises steeply with the number of factors present, and is minimal with none.
- Also higher with **T1 vs Ta**, **high grade**, **trigonal/periureteric location**, **vesicoureteric reflux**, and **CIS vs papillary disease** (including cystectomy for CIS vs for invasive cancer).
- Recurrences are usually **superficial and confined to the distal ureter** in some series, yet **more than half present with symptoms, most at advanced stage, 18–33% metastatic**; historical BCG-treated cohorts show high UTUC mortality.
- **Ureteric recurrence is commoner and earlier than pelvic recurrence.**
- **Risk is stable over long follow-up** → rigorous, prolonged upper tract surveillance.
- **BCG-refractory disease treated by cystectomy, and bladder CIS treated with BCG: ≈ 30% UTUC** → **annual upper tract evaluation** in these and all high-risk bladder cancers.
- Meta-analysis after cystectomy: **type of urinary diversion does not affect risk**; detection is usually symptom-driven; **CT should replace cytology and excretory urography** for follow-up.

Bladder recurrence after upper tract tumours.

- **Bladder recurrence 15–75% within 5 years** (22–47% in pooled contemporary data), **usually within 2 years**, often at the site of bladder trauma from ureterectomy.

- Explanations: downstream seeding, longer carcinogen dwell time, more urothelial cells. **Downstream seeding is most accepted** — bladder recurrences are **monoclonal**. **Low-grade UTUC** recurs less rapidly and shows **clonal discordance in nearly half**, supporting a **field effect**.
- Risk rises with **UTUC multifocality and higher stage**; **prior bladder tumour** was the only independent risk factor in one multivariable study.

PREDICTORS OF INTRAVESICAL RECURRENCE AFTER RNU (META-ANALYSIS; EAU)

Patient-specific	Tumour-specific	Treatment-specific
Male sex	Positive preoperative cytology	Laparoscopic approach
Previous bladder cancer	Ureteric location	Extravesical bladder cuff removal
Smoking	Multifocality	Positive surgical margins
Preoperative chronic kidney disease	Invasive pT stage	Diagnostic URS (especially pelvic tumours)
	Necrosis	

- **A single postoperative intravesical chemotherapy dose reduces bladder recurrence after RNU** (trials under way after URS) → **scheduled cystoscopic follow-up for all UTUC**.

Carcinoma in situ.

- **Bilateral disease 3–5% overall, ≈ 25% with CIS**; CIS implies multifocal and **pan-urothelial risk** → conservative management where feasible, with follow-up of **bladder, upper tracts and urethra**.

Metastatic Potential of Upper Urinary Tract Carcinoma

- **Non-organ-confined disease (≥ pT2 spread beyond the organ) is the strongest predictor of metastasis**, followed by **vascular** and then **lymphatic invasion**.

Route	Key facts
Direct	Renal parenchyma, surrounding structures; high grade → invasive, non-organ-confined disease
Lymphatic	Depends on depth and site . Pelvis and upper ureter → hilar, para-aortic, paracaval, interaortocaval ; lower ureter → ipsilateral common iliac and pelvic
Haematogenous	Liver, lung, bone ; direct extension into renal vein/vena cava is very rare
Epithelial	Seeding or intraepithelial migration → synchronous and metachronous tumours

Monoclonal (seeding) theory	Field-effect theory
Multiple tumours arise from one transformed clone , spreading antegrade (commoner) or retrograde	Mutagenic environment → independent de novo clones throughout the urothelium
Explains high bladder recurrence and recurrence in a retained ureteric stump after incomplete ureterectomy	Explains multifocal tumours of different clonal origin

- Current evidence: **both operate**.

Pan-urothelial disease.

- **Definition: bladder plus two extravesical sites** — in women, bladder and both upper tracts; in men, bladder with one or both upper tracts and/or the prostatic urethra.
- **Highest risk: high-risk multifocal non-muscle-invasive bladder tumours and CIS**. Management from small retrospective series: **cystectomy for high-grade or invasive bladder disease, conservative resection for non-infiltrating upper tract tumours**, radical excision for aggressive ones.
- Primary site (bladder vs upper tract) does not change outcome; some patients progress from **low-grade multifocal to high-grade invasive disease**, often with a personal or family history of cancer (genetic factors suspected).
- Sequential cystectomy then RNU: **more advanced pathology at RNU**, nodal disease predicts mortality, and **loss of renal function makes patients ineligible for cisplatin**.

- **No evidence-based algorithm; cystectomy for multifocal high-grade disease, close upper tract surveillance; total urinary tract removal** is an option in younger patients.

KEY POINTS: STAGING AND CLASSIFICATION

- Ureteric tumours: distal 70%, mid 25%, upper 5%. Bilateral disease 3–5%.
- Recurrence prognosis after RNU depends on tumour location and stage.
- Upper tract recurrence after bladder cancer 2–4%, over 17–170 months.
- Bladder recurrence after UTUC 15–75% within 5 years.
- CIS brings a higher incidence of bilateral and multifocal disease.
- Spread is typically by direct extension, lymphatics and bloodstream.

Concept — Recurrence is **asymmetric because urine flows downhill**: seeding runs antegrade, so bladder recurrence after UTUC is common and early while upper tract recurrence after bladder cancer is uncommon and late; the **field effect** explains the rest, and CIS is its most concentrated form.

PROGNOSIS

TNM Classification

- Classification and morphology mirror the bladder: **non-invasive papillary tumours** (papillary urothelial neoplasm of low malignant potential, low- and high-grade papillary carcinoma), **flat lesions (CIS)** and **invasive carcinoma**. **Variant histology is an adverse factor**.
- Regional nodes: **pelvis and upper ureter → hilar and retroperitoneal**; **mid and distal ureter → intrapelvic**. **Laterality does not affect N category**.
- Renal pelvic pT3 subclassification: pT3a = microscopic renal parenchymal infiltration; pT3b = macroscopic infiltration or peripelvic fat invasion (worse).

TABLE 138.1 TUMOUR NODE METASTASIS (TNM) CLASSIFICATION FOR UPPER URINARY TRACT CARCINOMA

From Brierley JD, et al.: *TNM classification of malignant tumours*, 8th edition, 2017.

Category	Definition
TX / T0	Cannot be assessed / no evidence of primary tumour
Ta	Non-invasive papillary carcinoma
Tis	Carcinoma in situ
T1	Invades subepithelial connective tissue
T2	Invades muscularis
T3	Beyond muscularis — pelvis: into peripelvic fat or renal parenchyma; ureter: into periureteric fat
T4	Adjacent organs, or through the kidney into perinephric fat
NX / N0	Cannot be assessed / no regional node metastasis
N1	Single node ≤ 2 cm
N2	Single node > 2 cm, or multiple nodes
M0 / M1	No distant metastasis / distant metastasis

- The table tests the **two sites of T3** (pelvic parenchyma counts as T3; **through the kidney into perinephric fat is T4**) and the **2 cm nodal cut-off**.
- **Trap**: renal parenchymal invasion by a pelvic tumour is **T3, not T4**.

TABLE 138.2 AMERICAN JOINT COMMITTEE ON CANCER (AJCC) STAGING SYSTEM IN CONJUNCTION WITH THE TNM SYSTEM

AJCC stage	TNM
0a / 0is	Ta N0 M0 / Tis N0 M0
I	T1 N0 M0
II	T2 N0 M0
III	T3 N0 M0
IV	T4 N0 M0, or any T with N1–N2, or any M1

- **Any nodal disease is stage IV** — the AJCC groups progress stage by stage with T, then jump to IV for T4, N+ or M1.

Histologic Grading

World Health Organization (WHO) 1973	WHO 2004/2016
G1, G2, G3	Papillary urothelial neoplasm of low malignant potential (PUNLMP), low grade, high grade
Used for decades	Basis of current EAU guidelines

Preoperative Factors

Factor	Effect
Age	Independent predictor of worse survival, yet RNU can cure the elderly — age alone must not preclude curative treatment
Sex	Not an independent prognostic factor
Race / ethnicity	Controversial: no difference in multicentre series; worse for Black patients in population data; Chinese and American patients differ in risk factors and disease features; Japanese vs European survival equal
Smoking	Raises mortality, disease recurrence and intravesical recurrence after RNU; cessation improves cancer control
Tumour location	Ureteric and/or multifocal worse than pelvic, stage-adjusted
Surgical delay	Invasive UTUC: RNU within 12 weeks of diagnosis and decision
Comorbidity and fitness	High American Society of Anesthesiologists (ASA) score, performance status (PS), Charlson Comorbidity Index → worse outcome
Body mass index	Obesity worsens cancer-specific outcome after RNU
Blood biomarkers	Locally advanced disease and cancer-specific mortality with low albumin, high neutrophil-to-lymphocyte ratio, adverse modified Glasgow score, high C-reactive protein, high De Ritis ratio (AST/ALT), high fibrinogen, impaired renal function

Postoperative Factors

Tumour Stage and Grade

- **Stage and grade are the primary prognostic factors; stage is the single most important.** Muscle-invasive UTUC does badly.

FIVE-YEAR CANCER-SPECIFIC SURVIVAL BY STAGE

Series	Non-muscle-invasive / T1N0	pT2	pT3	pT4 or N+
Older series	—	< 50% (pT2/pT3)		< 10% (pT4)
Contemporary cohort	86%	70% organ-confined muscle-invasive; 44% locally advanced		—
SEER, high-risk RNU	86%	77%	63%	39%

- **Why the figures differ:** different eras, case mix and categories (pooled pT2/pT3 vs separated; organ-confined vs locally advanced). The contemporary pattern is a **stepwise fall, steepest beyond pT3**.
- **pT3a vs pT3b** is prognostically relevant but **not in routine use**.
- **Figure 138.5** (main prognostic factors, after EAU 2017): grouped as pre- and postoperative factors, including ASA score, body mass index and Eastern Cooperative Oncology Group performance status (ECOG-PS).

Architecture, grade and histology

- **Benign papillomas do well regardless of treatment.** PUNLMP — reassigned from 1973 grade 1 to the low-grade category — is **managed as low-grade disease**.
- **Papillary tumours do better than sessile; sessile architecture is a strong predictor of poor outcome.**
- **Renal pelvic tumours are slightly commoner than ureteric;** after conservative treatment, recurrence is **antegrade (downstream) in 33–55%, proximal recurrence uncommon**.
- **Atypia, dysplasia or CIS in 60–95% of RNU specimens** for pelvic tumours; **concomitant CIS in organ-confined UTUC and prior bladder CIS raise recurrence and cancer-specific mortality**.

Size and necrosis

- **Size is not an established criterion**, but > 3–4 cm is linked to worse survival and bladder recurrence; each additional centimetre raises the risk of ≥ pT2; **2 cm is the best cut-off** for predicting ≥ pT2 at RNU.
- **Extensive necrosis (> 10% of tumour area) is an independent predictor** after RNU, tracking advanced stage, high grade, nodal disease, CIS and sessile architecture; its effect on survival is debated.

History and multifocality

- **Metachronous bilateral UTUC is rare (≈ 3%),** occurs in the elderly and carries short survival.
- **History of bladder tumour in 15–50% of UTUC; CIS of the distal ureter in 7–25% of cystectomy specimens;** upper tract recurrence after bladder cancer **2–4% (mean interval ≈ 70 months), 3–9% after cystectomy**.
- **Environmental, analgesic and LS-related UTUC recur more often multiply and bilaterally.** Balkan tumours are usually low grade, but coexisting renal insufficiency limits kidney-sparing options. Papillary scarring from phenacetin predicts higher grade and progression, dose-dependently.

Pattern of spread

- **Most metastases appear within 2–3 years; sites: lung, liver, bone, regional nodes.**
- **Stage T3 renal pelvic tumours outlive stage T3 ureteric tumours** — the renal parenchyma acts as a barrier, whereas periureteric spread enters a rich vascular and lymphatic plexus; ureteric tumours have higher local and distant failure at equal stage and grade.

Lymph Node Involvement

- **Node metastasis and extranodal extension** are powerful, independent predictors of poor survival; **up to 40% harbour nodal disease** depending on stage and grade.

- **Lymph node dissection (LND) at RNU gives optimal staging; its curative role is debated. Lymph node density $\geq 30\%$ predicts poor outcome.**
- **Clinically node-positive UTUC $\rightarrow$ systemic chemotherapy first, surgery if there is a response.**
- **Variant histology** (commonest: squamous, micropapillary, sarcomatoid) $\rightarrow$ locally advanced disease and poor cancer-specific and overall survival.

Lymphovascular Invasion

- **Lymphovascular invasion (LVI) in $\approx 20\%$; independent predictor of recurrence and survival in node-negative or node-unknown patients, not in node-positive patients.**
- **Not identifiable on URS biopsy** — a postoperative factor only; **must be reported on every UTUC specimen.**

Surgical Margins

- **Positive soft-tissue margin after RNU predicts recurrence; above pT2, pathologists should report margins at the ureteric transection, bladder cuff and around the tumour.**

Other Pathological Factors

- **Extensive necrosis ($> 10\%$); pathological downstaging after neoadjuvant chemotherapy $\rightarrow$ better overall survival; sessile architecture; bladder CIS history and concomitant CIS; peripelvic fat or macroscopic invasion (pT3b) worse than microscopic parenchymal invasion (pT3a).**

Molecular Markers

- UTUC and bladder cancer share some genetic and epigenetic features but are **divergent entities**. Evidence is retrospective and small: **currently insufficient for routine clinical use** of any marker, though some are promising.
- **Microsatellite instability (MSI) and hypermethylation** are the key differences from bladder cancer. **MSI is an independent prognostic marker**, flags **germline mutations (LS)**, and is linked to **inverted growth pattern**; MSI testing on tumour and normal tissue, established for colon cancer, can be applied to UTUC meeting LS criteria. A urine MSI panel detected most UTUCs.
- **Chromosome 9 alterations in $\approx 50\%$** . Comparative genomic hybridisation shows **concordant losses (2q, 8p, 9q, 11p, 13q, 17p, 18q) and gains (1q, 6p, 8q, 17q)** between pelvic and bladder tumours, with similar gene expression. Other alterations: +1p36, +6p22, +7, +8q22, -9p21, +11q, -13q, +17, +19q13, +20q.
- **PD-1 and PD-L1 expression** is prognostic in high-grade UTUC.

TISSUE MARKERS AND THEIR PROGNOSTIC DIRECTION

Pathway	Marker	Association
Cell cycle	TP53	Positive staining on URS cytology strongly predicts UTUC; negative staining excluded it. Altered expression linked to stage and prognosis, not independent of stage and grade
Cell cycle	CDKN1B (p27)	Low staining $\rightarrow$ worse disease-specific survival
Apoptosis	Bcl-2, survivin	Advanced cancer; survivin $\rightarrow$ disease-specific survival
Migration and invasion	E-cadherin, matrix metalloproteinases	Poor prognosis; metalloproteinases track pT stage
Angiogenesis	Hypoxia-inducible factor-1α (HIF-1α)	Expressed in two-thirds of UTUC, absent in normal urothelium; high T, N, grade, worse cancer-specific survival
Proliferation	Ki-67	Progression, worse survival, metachronous tumours
Proliferation		

Pathway	Marker	Association
	Epidermal growth factor receptor (EGFR); nuclear factor-κB; HER2	EGFR: stage, grade, squamous differentiation. NF-κB: predicts disease-specific and overall survival. HER2 overexpression rare ; higher stage and grade
Differentiation	Uroplakin III	Expression is favourable (lower stage and grade); outperformed stage and nodal status on multivariable analysis
Differentiation / EMT	Snail	Strong in invasive UTUC; stage, grade, LVI; predicts recurrence and survival
Telomerase	hTR	Positive in almost all UTUC; rises with stage
Mitosis	Aurora-A	Vascular invasion, recurrence
MET family	c-MET vs RON	c-MET: vascular invasion, worse outcome. RON: no correlation
Inflammation	Cyclooxygenase-2 (stromal)	High stage and grade, poor prognosis

- **Pattern:** nearly every marker of proliferation, invasion or hypoxia is adverse; the **exceptions to remember** are **uroplakin III** and **CDKN1B**, where **expression (not loss)** is favourable, and **RON**, which **does not correlate**.

Other Markers

Test	Bottom line
DNA ploidy (flow cytometry)	Aneuploidy → poor long-term survival
NMP22 (nuclear matrix protein)	Elevated in UTUC; more sensitive, less specific than cytology
Urine FISH (UroVysion)	Much more sensitive than cytology , slightly less specific; in haematuria with negative cytology, a negative FISH effectively excluded UTUC
Fibrinogen–fibrin degradation products (AuraTek FDP)	More accurate than bladder tumour antigen (BTA) test and cytology

Predictive Tools

- **Two preoperative models:** one predicts **nodal involvement** (guiding whether and how extensively to do LND at RNU); one identifies **non-organ-confined disease** for RNU.
- **Five postoperative nomograms** use standard pathology; one uses **only four variables** with high accuracy for high-grade UTUC after surgery. None is yet established in practice.

Risk Stratification

- Clinical staging is unreliable, so **pre-RNU models predict ≥ pT2/non-organ-confined disease** to select patients for **neoadjuvant chemotherapy**; pre- and postoperative nomograms aid counselling, follow-up and **perioperative chemotherapy decisions**.
- Kidney-sparing selection uses **low- vs high-risk stratification** (Figure 138.6).

FIGURE 138.6 RISK STRATIFICATION OF UTUC

Modified from the EAU guidelines on upper urinary tract urothelial carcinoma, 2020 update.

Low-risk UTUC — all must be present	High-risk UTUC — any one suffices
Unifocal disease	Multifocal disease
Tumour < 2 cm	Tumour > 2 cm
Low-grade cytology	High-grade cytology
Low-grade URS biopsy	High-grade URS biopsy
No invasive aspect on CT urography	Hydronephrosis

Low-risk UTUC — all must be present	High-risk UTUC — any one suffices
—	Previous radical cystectomy for bladder cancer
—	Variant histology

- The figure is testing the **asymmetry of the rule**: low risk requires **every** favourable feature; a **single** adverse feature makes the tumour high risk.
- **Trap**: a small, low-grade unifocal tumour **with hydronephrosis is high risk**.

KEY POINTS: PROGNOSTIC FACTORS

- Use the TNM classification and staging system.
- Use the WHO 2004/2016 grading classification.
- Age alone should not prevent a curative approach.
- Ureteric and/or multifocal tumours have a worse prognosis than renal pelvic tumours.
- Muscle-invasive tumours have a poor prognosis.
- Positive regional nodes carry a poor survival prognosis.
- Lymphovascular invasion predicts recurrence and survival.

Concept — *Prognosis is stage first, then everything that stands in for stage: because true stage is known only after RNU, the preoperative factors (grade, size, hydronephrosis, location, multifocality) are all surrogates for invasion — and the risk-stratification rule simply asks whether any surrogate points to it.*